Technische Universität Dortmund

Seminar Soccer Analytics

A Reassessment of Penalty Shootout Sequences: Disproving the Findings of Csató and Petróczy(2021)

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1 Introduction

In high-stakes soccer competitions, penalty shootouts serve as the decisive mechanism to determine the winner when matches remain tied after regular and extra time. The fairness of these shootouts has been a subject of considerable debate, particularly due to the **First Mover Advantage (FMA)**, which suggests that the team shooting first (Team A) may have a systematically higher probability of winning than the team shooting second (Team B). The primary reason for this imbalance is the psychological pressure associated with kicking second, where players often experience greater anxiety and lower scoring probabilities. A significant contribution to this discussion is the work by László Csató and Dóra Gréta Petróczy in the paper "Fairness in penalty shootouts: Is it worth using dynamic sequences?", which evaluates the fairness of different penalty shootout mechanisms, including both static and dynamic shooting sequences. Their study models the probability of scoring penalties under the assumption that the first kicker scores with probability p, while the second kicker scores with a lower probability q, where $q \leq p$ due to psychological pressure. The paper introduces mathematical formulas to calculate the probability of Team A winning for each sequence and examines the fairness of various mechanisms under different values of p and q.

1.1 Summary of the Main Paper and Its Findings

The study by Csató et al. (2021) is structured into several key sections. It first defines the problem by evaluating the fairness of penalty shootout mechanisms through a probabilistic framework. The main objective is to determine whether dynamic rules, which adjust the shooting order based on the score, offer fairness advantages over static designs like the traditional **ABAB** sequence.

The authors introduce a mathematical model to quantify the First Mover Advantage (FMA). The central assumption is that Team A (first shooter) has a scoring probability p, whereas Team B (second shooter) has a lower probability $q \leq p$. Given these probabilities, they derive explicit formulas for the winning probability of Team A under different penalty shootout formats. The penalty sequence is deemed to be fair when the probability of winning the shootout is **0.5** for both teams. Several penalty shootout mechanisms are analyzed, categorized into static and dynamic designs. Static sequences follow a fixed order throughout the shootout. The Standard (**ABAB**) rule is the official format, where Team A always kicks first, followed by Team B. However, this design has been criticized for favoring the first-kicking team. The Alternating (**ABBA**) rule attempts to mitigate this by reversing the order every two kicks, ensuring that both teams experience the advantage and disadvantage of kicking first. An extension of this concept is the **ABBA|BAAB** rule, which mirrors the sequence every four kicks, alternating between ABBA and BAAB. This design is inspired by the Thue–Morse (TM) sequence, which theoretically improves fairness by balancing first-mover effects over a longer sequence. Dynamic sequences, on the other hand, adjust the order based on the score at each stage of the shootout. The **Catch-up** rule ensures that the first kicker alternates unless the previous round ended with the first kicker missing and the second succeeding, in which case the order remains unchanged. A modification of this is the **Adjusted Catch-up** rule, which follows the Catch-up rule for the first five rounds but guarantees that Team B always kicks first in sudden death. The **Behind-first** rule dictates that the team trailing in score takes the first kick in the next round. If the score is tied, the order of the previous round is mirrored. The **Adjusted Behind-first** rule follows the same principle but ensures that Team B always kicks first in sudden death. The authors formulate the probability of Team A winning under these mechanisms. For the standard ABAB rule, the probability of Team A winning in the sudden death stage is given by

$$W^S(A) = \frac{p(1-q)}{2p - pq - p^2}.$$

For the ABBA rule and dynamic mechanisms, the probability of Team A winning is expressed as:

$$W^R(A) = \frac{1 - p + p^2}{2 - 2p + pq + p^2}.$$

The ABBA|BAAB rule, which is considered the fairest in the study, uses a more complex formula:

$$W^{AA}(A) = \frac{p(1-q) + (1 - 2p + pq + p^2)p(1-q) + (1 - 2p + pq + p^2)^2}{1 + (1 - 2p + pq + p^2)^2}.$$

$$W^{AB}(A) = \frac{p(1-q) + (1 - 2p + pq + p^2)(1-p)p + (1 - 2p + pq + p^2)^2}{1 + (1 - 2p + pq + p^2)^2}.$$

The winning probabilities for Team A under different penalty shootout mechanisms are shown in Figure 1 and Table 1.

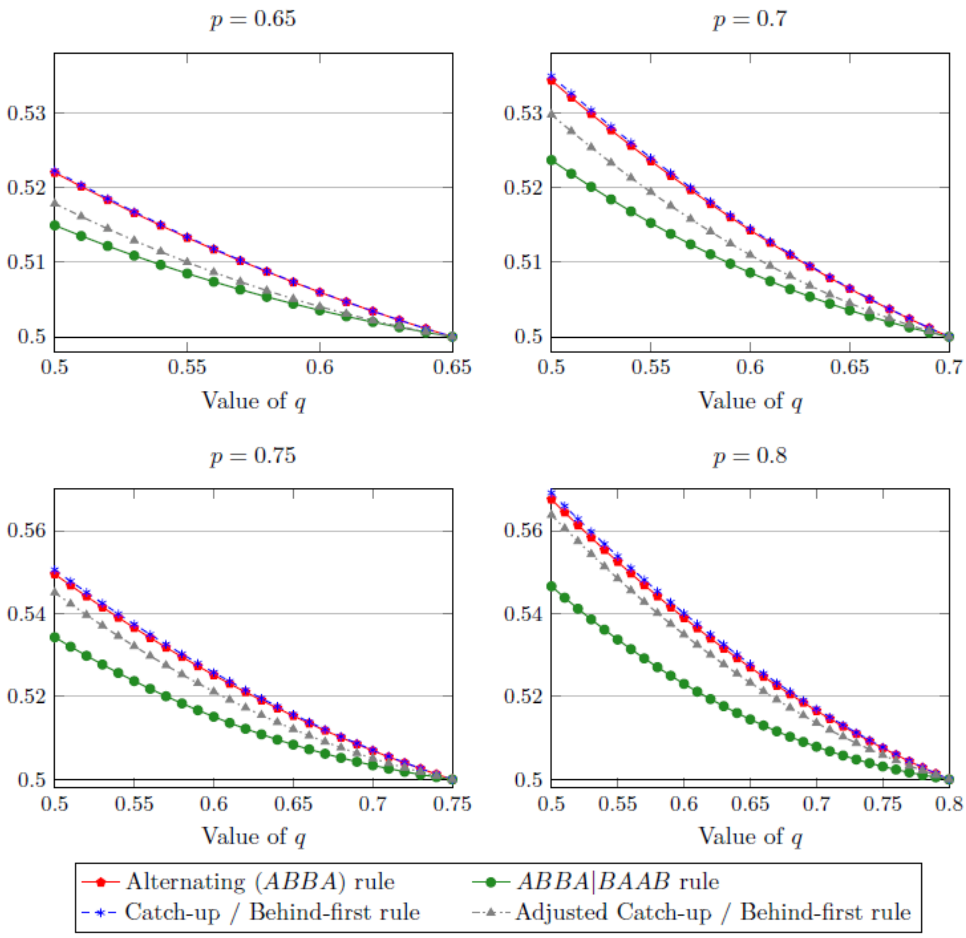

Figure 1: The probability that team A wins a penalty shootout over five rounds including the sudden death stage.

Figure 1 presents the probability that Team A wins for different values of p and q. The results confirm that fairness is difficult to achieve because adjusting the first kicker's scoring probability cannot fully balance the disadvantage of Team B. The ABBA rule performs comparably to dynamic sequences, making their added complexity unnecessary. Adjustments to dynamic rules, such as guaranteeing the first penalty in sudden death for Team B, improve fairness but have diminishing effects as p increases. The ABBA|BAAB rule is the closest to fairness and, if ABBA is deemed inadequate, moving towards Thue–Morse sequences may offer further fairness improvements. The results in Table 1 further indicate that the ABAB rule is the most unfair, consistently favoring Team A with a 57.14% win rate. The ABBA rule reduces this bias slightly but remains imperfect. The ABBA|BAAB rule performs the best, bringing the probability closer to 50%. Dynamic mechanisms such as Catch-up and Behind-first fail to offer improvements.

Table 1: Winning Probabilities for Team A in each shootout round, for $(p = \frac{3}{4}, q = \frac{2}{3})$

Mechanism	1	2	3	4	5	6	7	8
ABAB	57.14	57.14	57.14	57.14	57.14	57.14	57.14	57.14
ABBA	52.00	51.62	51.45	51.38	51.24	51.23	51.12	51.14
ABBA|BAAB	51.04	50.44	50.74	51.00	50.66	50.41	50.65	50.80
Catch-up	52.00	51.50	51.49	51.30	51.27	51.18	51.15	51.10
Adjusted Catch-up	52.00	50.07	51.25	50.58	50.96	50.73	50.85	50.77
Behind-first	52.00	51.50	51.49	51.30	51.27	51.18	51.15	51.10
Adjusted Behind-first	52.00	50.07	51.25	50.58	50.96	50.73	50.85	50.77

2 Problem Statement

This section presents the problem statement and describes the datasets used in carrzing out the analysis.

2.1 Identification of Research Problem and Approach

The original study conducted by Csató and Petróczy provides a theoretical framework for evaluating fairness in penalty shootouts by comparing static and dynamic kicking sequences. However, this study has significant limitations, primarily due to unrealistic assumptions about fixed probabilities p and q, reflecting the scoring probability of the first and second shooters, respectively. These probabilities remain constant throughout the entire shootout, ignoring potential psychological dynamics and the evolving state of the match. Consequently, the analysis does not incorporate empirical data or dynamic updating methods, significantly restricting the practical applicability of the conclusions.

The second paper, titled "Ordering sequential competitions to reduce order relevance: Soccer penalty shootouts", authored by Nils Rudi et al., directly addresses these issues by employing extensive empirical datasets and advanced bootstrapping methods to simulate and assess fairness across different penalty shooting sequences (**ABAB**, **ABBA**, and **ABBA|BAAB**) more realistically. However, the second paper does not include analyses for dynamic shooting sequences, leaving an important research gap.

To bridge this gap, we extend the analysis performed in Rudi et al. (2021) by incorporating dynamic penalty shootout sequences (**Catch-up**, **Adjusted Catch-up**, **Behind-first**, **Adjusted Behind-first**). Our methodological approach begins with the bootstrapping framework originally implemented by Rudi et al. for static sequences, leveraging the same empirical penalty data to ensure consistency and comparability of results. Subsequently, we develop new R-code to simulate dynamic sequences using a Bayesian updating framework, dynamically estimating scoring probabilities after each kick. Our simulations use Bayesian inference with beta-binomial priors, adapting to observed outcomes throughout each shootout and thus overcoming the limitations of fixed probabilities used in the original Csató study.

2.2 Data Description and Quality

The empirical analysis relies on several datasets. The primary dataset, `shootout_data.csv`, contains detailed penalty shootout records. It includes the following variables: `uri` (discrete nominal identifier), `competition` (nominal categorical variable identifying the tournament), `date` (date of match, discrete temporal variable), `home` and `away` (nominal categorical variables identifying the competing teams), `winner` (nominal categorical variable), and `goal_scored` (nominal dichotomous outcomes, "Y" for success and "N" for failure). Additionally, from this raw data, metric discrete variables such as `kickNumber` (penalty kick sequence), `round` (shootout round), `teamAScore` and `teamBScore` (cumulative discrete metric scores), and `gd` (goal difference, discrete integer) were derived.

The auxiliary dataset `aph_competition_mapping.csv` consists of discrete nominal variables, including `competition_id` (numerical identifier), `competition_name` (nominal categorical variable describing the competition), and `aph_id` (numerical identifier aligning with data used in the APH paper). Another auxiliary dataset, `international_competitions.csv`, includes nominal dichotomous variables: `competition_id` (numerical identifier) and `type` (categorical classification variable with two levels, "International" or "Club", distinguishing the tournament type).

Furthermore, the bootstrapped datasets used from prior simulations, including `aph_ABAB.csv`, `aph_ABBA.csv`, `aph_tm.csv`, `extended_ABAB.csv`, `extended_ABBA.csv`, and `extended_tm-1.csv`, each contain numeric variables named `val`, representing simulated winning probabilities for Team A under various static penalty sequences. These numeric variables are continuous metric variables used directly in Bayesian updating processes as empirical inputs for prior probability estimation.

The **APH** datasets (`aph_ABAB.csv`, `aph_ABBA.csv`, `aph_tm.csv`) refer to a subset of penalty shootout data aligned with the study conducted by **Apesteguia and Palacios-Huerta (2010)**. These datasets contain penalty shootout simulations that match the competitions and methodology used in their research, allowing for direct comparisons with previous studies on penalty shootout fairness. On the other hand, the **Extended** datasets (`extended_ABAB.csv`, `extended_ABBA.csv`, `extended_tm-1.csv`) represent a broader and more comprehensive dataset, incorporating additional penalty shootouts beyond those covered in the APH dataset. These datasets include penalty shootouts from **more competitions, a broader timeframe, or additional contextual factors** not present in the APH data. The purpose of the Extended datasets is to provide a **larger empirical basis** for evaluating penalty shootout fairness beyond the scope of prior research.

All datasets used in this study were obtained from **Olivares (2020)**, which compiled penalty shootout records from multiple football competitions worldwide. The data source includes both **historical records and structured shootout simulations**, ensuring a robust foundation for empirical analysis. A preliminary data quality assessment was conducted using R-code to identify potential missing values (NAs). The results indicate no presence of missing values across any variables in all datasets.

3 Statistical Methods

This section presents the different statistical methods and the R Software used to assess penalty shootout fairness.

3.1 Bayesian Updating using the Beta-Binomial Model

The statistical framework utilized in this analysis relies on Bayesian updating, specifically employing the Beta-Binomial conjugate prior model (Murphy et al., 2012, Page 72-78). Bayesian inference provides a systematic method for updating the probability of a hypothesis given observed data. This approach combines prior beliefs about parameters with evidence from observed data, resulting in updated beliefs represented by the posterior distribution.

The Beta-Binomial model is particularly useful for modeling scenarios involving binary outcomes. Consider an experiment consisting of a series of Bernoulli trials, each yielding either a success (with probability θ) or failure (with probability $1-\theta$). Given a set of observations $\mathcal{D} = \{x_1, x_2, \ldots, x_N\}$, where x_i takes the value 1 for success and 0 for failure, the likelihood function for the parameter θ is defined as:

$$p(\mathcal{D}|\theta) = \theta^{N_1}(1-\theta)^{N_0},$$

where $N_1 = \sum_{i=1}^{N} x_i$ is the number of successes, and $N_0 = N - N_1$ is the number of failures.

In Bayesian inference, the parameter θ is treated as a random variable with a prior distribution, representing the initial belief about its possible values before observing the data. The conjugate prior for the Bernoulli likelihood is the Beta distribution, which has the form:

$$p(\theta|\alpha,\beta) = \frac{\theta^{\alpha-1}(1-\theta)^{\beta-1}}{B(\alpha,\beta)},$$

where $B(\alpha, \beta)$ is the Beta function defined as:

$$B(\alpha,\beta) = \frac{\Gamma(\alpha)\Gamma(\beta)}{\Gamma(\alpha+\beta)},$$

and $\Gamma(\cdot)$ denotes the Gamma function. Here, α and β are hyperparameters of the Beta distribution, which represent the prior belief about the number of successes and failures, respectively. The Beta distribution is used as a conjugate prior because multiplying a Beta prior with a Binomial likelihood results in a posterior that is also Beta-distributed. This property simplifies Bayesian updating, as the posterior has the same functional form as the prior. Formally, the posterior distribution of θ is computed using Bayes' theorem:

$$p(\theta|\mathcal{D},\alpha,\beta) \propto p(\mathcal{D}|\theta)p(\theta|\alpha,\beta).$$

Due to the conjugacy of the Beta and Binomial distributions, the posterior distribution also follows a Beta distribution. Specifically, the posterior parameters are updated as follows:

$$p(\theta|\mathcal{D},\alpha,\beta) = \mathrm{Beta}(\theta|\alpha+N_1, \beta+N_0).$$

The parameters α and β can be interpreted as pseudo-counts of prior successes and failures before any data is observed. If no strong prior knowledge exists, a common choice is $\alpha = \beta = 1$, which corresponds to a uniform (uninformative) prior. Alternatively, when prior information is available, these hyperparameters can be chosen to reflect expert knowledge or historical data. The posterior mean, which provides a point estimate of the updated parameter θ, is then calculated as:

$$\mathbb{E}[\theta|\mathcal{D}] = \frac{\alpha + N_1}{\alpha + \beta + N}.$$

This posterior mean represents the updated estimate of θ given prior knowledge and observed data. It is essentially a weighted average of the prior belief and the empirical mean from the observed data. A higher number of observations (N) reduces the influence of the prior, shifting the posterior closer to the empirical estimate.

3.2 Bootstrapping

Bootstrapping (Davidson , A.C. et al., 1997, Page 11-21) is a resampling technique used to approximate the distribution of a statistic by generating multiple synthetic datasets from the original data. It is a non-parametric method that does not rely on strong distributional assumptions, making it particularly useful for estimating standard errors, confidence intervals, and hypothesis testing in complex statistical problems.

Given a dataset $X = \{x_1, x_2, \ldots, x_n\}$ of observed values, the bootstrap procedure involves constructing B resampled datasets X_b^* by drawing observations *with replacement* from the original data:

$$X_b^* = \{x_1^*, x_2^*, \ldots, x_n^*\}, \quad \text{where } x_i^* \sim X.$$

Each resampled dataset maintains the same size as the original data (n) but introduces variability by including duplicate observations while potentially omitting others. This resampling captures the inherent randomness in the original dataset. For each bootstrap sample, a statistic of interest $\hat{\theta}_b^*$ (e.g., the mean, median, variance) is computed:

$$\hat{\theta}_b^* = g(X_b^*),$$

where $g(\cdot)$ represents the function applied to the resampled dataset. The empirical distribution of $\hat{\theta}^*$ across all B samples approximates the true sampling distribution of $\hat{\theta}$.

The standard error of an estimator $\hat{\theta}$ is a key measure of its variability. In classical inference, this is often estimated using parametric formulas. The bootstrap method provides a data-driven alternative:

$$SE^*(\hat{\theta}) = \sqrt{\frac{1}{B}\sum_{b=1}^{B}\left(\hat{\theta}_b^* - \bar{\theta}^*\right)^2},$$

where $\bar{\theta}^* = \frac{1}{B}\sum_{b=1}^{B}\hat{\theta}_b^*$ is the mean of the bootstrapped estimates.

The bootstrap method is frequently used to construct confidence intervals for an unknown parameter θ. The two most commonly used bootstrap confidence intervals are:

- **Percentile Interval**: This method directly uses quantiles from the bootstrap distribution. For a confidence level $1-\alpha$, the interval is given by:

$$[\hat{\theta}^*_{\alpha/2}, \hat{\theta}^*_{1-\alpha/2}],$$

  where $\hat{\theta}^*_{\alpha/2}$ and $\hat{\theta}^*_{1-\alpha/2}$ are the empirical quantiles of the bootstrap estimates.

- **Bias-Corrected and Accelerated (BCa) Interval**: This method adjusts for bias and skewness in the bootstrap distribution and is given by:

$$[\hat{\theta}^*_{z_1}, \hat{\theta}^*_{z_2}],$$

  where z_1 and z_2 are adjusted quantiles incorporating bias and acceleration corrections.

Bootstrapping can be implemented in two primary ways:

- **Non-Parametric Bootstrapping**: Resamples are drawn directly from the empirical distribution of the observed data. No assumption is made about the underlying distribution.

- **Parametric Bootstrapping**: Instead of directly resampling, data is generated from a fitted parametric model. This is useful when prior knowledge about the distribution is available.

3.2.1 Monte Carlo Implementation of Bootstrapping

In practical applications, bootstrapping is often implemented as a Monte Carlo simulation. The procedure consists of the following steps: 1. Resample B bootstrap datasets X_b^* from the original data. 2. Compute the statistic $\hat{\theta}_b^*$ for each resampled dataset. 3.

Aggregate the results to estimate the empirical distribution of $\hat{\theta}$. 4. Compute derived quantities such as standard errors or confidence intervals. This function generates B resampled datasets and computes the desired statistic across all iterations.

3.3 Markov Chains

A **Markov Chain** (Levin, D. A. et al., 2006, Page 2-7). Markov chains and mixing times. American Mathematical Society.) is a stochastic process that models transitions between different states within a system where the probability of moving to the next state depends only on the current state and not on the sequence of previous states. This property is known as the **Markov Property** and can be mathematically expressed as:

$$P(X_{t+1} = y | X_t = x, X_{t-1} = x_{t-1}, ..., X_0 = x_0) = P(X_{t+1} = y | X_t = x)$$

for all states x, y in the state space $\mathcal{X}$. This equation implies that the future state X_{t+1} depends only on the present state X_t and not on the path taken to reach X_t. A Markov Chain is characterized by a **transition matrix** P, which defines the probabilities of transitioning from one state to another. The transition probability from state x to state y is denoted as:

$$P(x, y) = P(X_{t+1} = y | X_t = x)$$

where the entries of P satisfy the stochastic property:

$$\sum_{y \in \mathcal{X}} P(x, y) = 1, \quad \forall x \in \mathcal{X}$$

indicating that the sum of transition probabilities from any state must equal 1. Consider a simple Markov Chain example where a frog jumps between two lily pads, denoted as e (east) and w (west). The frog flips a biased coin at each step to determine whether to stay on its current lily pad or jump to the other. The transition probabilities can be represented by the following transition matrix:

$$P = \begin{bmatrix} 1-p & p \\ q & 1-q \end{bmatrix}$$

where:

- p is the probability of jumping from e to w,
- q is the probability of jumping from w to e.

A key aspect of Markov Chains is determining their **stationary distribution**, which describes the long-term probabilities of being in each state. The stationary distribution π satisfies:

$$\pi P = \pi$$

which means that applying the transition matrix to π leaves it unchanged. For the frog example, the stationary probabilities of being in states e and w are given by:

$$\pi(e) = \frac{q}{p+q}, \quad \pi(w) = \frac{p}{p+q}$$

indicating that as time progresses, the probability of finding the frog on a particular lily pad converges to these values, regardless of the starting state.

3.4 Software Tools

The analyses were conducted using **R** (R Core Team, 2024), version 4.4.2. Data processing was performed with `dplyr` (Wickham et al., 2023) to manipulate and summarize data efficiently. The `plyr` package (Wickham, 2011) was employed for structured data transformation using the Split-Apply-Combine approach. SQL-like queries were executed within R using `sqldf` (Grothendieck, 2017).

For the simulation of penalty sequences, the `markovchain` package (Spedicato, 2017) was utilized to model probability distributions under Markovian assumptions. Bootstrapping was implemented using a custom R function that draws samples with replacement, aligning with principles outlined in Davison and Hinkley (1997).

Bayesian updating was performed using the Beta-Binomial model, with parameter estimation conducted via base R functions. Specifically, the `rbeta` function was used for drawing posterior probability estimates based on dynamically updated priors.

4 Results and Discussion

In this chapter, the statistical analysis, results and its interpretations will be presented.

4.1 Dataset and Analysis Framework

The empirical analysis is based on multiple datasets, including observed penalty shootout data and bootstrapped samples derived from statistical resampling. The primary dataset, `shootout_data.csv`, contains shot-by-shot penalty outcomes from historical matches, detailing the scoring sequence, competition type, and the final result. Each observation includes discrete and nominal dichotomous variables, such as the goal outcome (`"Y"` for success and `"N"` for failure), as well as discrete metric variables like `kickNumber`, representing the order of kicks, and round-by-round cumulative scores.

Additional datasets, such as `aph_competition_mapping.csv` and `international_competitions.csv`, provide metadata for classifying shootouts based on competition level (international vs. club tournaments). The bootstrapped datasets (`aph_ABAB.csv`, `aph_ABBA.csv`, `aph_tm.csv`, `extended_ABAB.csv`, `extended_ABBA.csv`, and `extended_tm-1.csv`) were generated using a resampling approach to create synthetic distributions of penalty shootout outcomes under different rule sets.

The methodology builds upon the bootstrapping techniques used in Rudi et al. (2021), who focused exclusively on static sequences. While Rudi et al. employed bootstrapping to evaluate fairness in ABAB and ABBA sequences, this study extends their framework by implementing Bayesian updating to model dynamic sequences. Specifically, I introduces **Bayesian updating with a Beta-Binomial model**, dynamically adjusting scoring probabilities based on observed outcomes. This method enables real-time estimation of scoring likelihoods, a key innovation that distinguishes this approach from previous research.

4.2 Application of Markov Chains in Penalty Shootouts

In the context of penalty shootouts, the process can be effectively modeled as a Markov Chain, where each penalty kick represents a transition between different game states. The state space consists of all possible score differences between the two teams, while the transition probabilities between these states are determined by empirical scoring rates.

The initial state begins with a score difference of zero before the first penalty is taken, and the process progresses as each shot is either scored or missed, altering the game state accordingly. The absorbing states represent the end of the shootout, where either team has won or all penalty attempts have been exhausted.

To model this process effectively, the probability of scoring a penalty is not assumed to be fixed but instead updated dynamically as the shootout progresses. This is achieved through Bayesian updating with a Beta-Binomial model, allowing the estimation of scoring probabilities to evolve based on observed outcomes. This dynamic updating method contrasts with previous approaches that assume a constant scoring likelihood throughout the shootout. By simulating thousands of penalty shootouts under different rules, the Markov Chain model allows us to analyze how various penalty shootout sequences influence fairness. The framework enables a deeper understanding of how the advantage or disadvantage of shooting first evolves over multiple rounds and how alternative penalty sequences, such as ABBA or Catch-Up, alter the distribution of winning probabilities.

4.3 Empirical Findings

The results of the analysis, summarized in Table 2, show clear differences between static and dynamic sequences. As expected, the ABAB rule exhibits a strong bias in favor of Team A, with a winning probability of **53.9%**. This confirms previous findings that ABAB unfairly advantages the first-shooting team. The ABBA sequence, designed to mitigate First Mover Advantage (FMA), reduces the bias significantly, yielding a more balanced **51.9%** win rate for Team A.

Table 2: Winning Probabilities for Team A under Different Penalty Shootout Formats

Shootout Format	Mean	1%	50% (Median)	99%
ABAB	0.5389	0.4916	0.5453	0.5572
ABBA	0.5196	0.4735	0.5271	0.5492
Thue-Morse (TM)	0.4742	0.4485	0.4737	0.4959
Extended ABAB	0.5482	0.5316	0.5468	0.5616
Extended ABBA	0.5095	0.4897	0.5106	0.5223
Extended TM	0.4837	0.4722	0.4822	0.4972
Catch-Up	0.4334	0.4059	0.4312	0.4431
Adjusted Catch-Up	0.4338	0.4267	0.4310	0.4608
Behind-First	0.4353	0.3971	0.4351	0.4392
Adjusted Behind-First	0.4366	0.4154	0.4378	0.4741

However, the results for dynamic sequences reveal a **major deviation** from the theoretical expectations outlined in Csató et al. (2021). Contrary to their assertion that Catch-Up and Behind-First rules promote fairness, the empirical findings demonstrate that these mechanisms **significantly bias the shootout in favor of Team B**. The Catch-Up and Behind-First sequences result in Team A win probabilities as low as **43.3–43.8%**, strongly disadvantaging the first shooter. Adjusted versions of these sequences also exhibit the same issue, with similar or slightly worse biases.

These results confirm that dynamic sequences introduce an unintended imbalance, systematically favoring the second shooter rather than creating fairness.

4.4 Comparison with Csató et al. (2021)

The results from my analysis directly contradict the fairness claims made in Csató et al. (2021). Their study suggested that dynamic sequences could counteract First Mover Advantage by adjusting the order of shooters based on the evolving score. They hypothesized that by giving the trailing team the first shot in later rounds, the pressure balance would be restored, leading to a more equitable shootout process.

However, the Bayesian simulation results reveal that the opposite occurs: **dynamic sequences disproportionately benefit Team B, leading to a structural disadvantage for Team A**. This deviation arises because Catch-Up and Behind-First rules frequently allow Team B to take critical shots in lower-pressure conditions. Instead of balancing psychological effects, these sequences **overcorrect** for First Mover Advantage, making them inherently unfair.

By contrast, ABBA remains the most effective balancing mechanism. The results confirm that **ABBA brings Team A's win probability close to 50%**, whereas dynamic rules create new distortions. This suggests that **Csató et al.'s conclusions were based on theoretical assumptions that do not hold under empirical testing**.

Furthermore, while Csató et al. (2021) claimed that **ABBA|BAAB (Thue-Morse) is the fairest**, my findings suggest otherwise. The Thue-Morse sequence does reduce the ABAB bias, but its win probability for Team A (47.4%) is significantly below 50%, suggesting that it **introduces a slight bias in favor of Team B rather than achieving fairness**. Therefore, **ABBA remains the superior alternative**, providing the best balance without additional complexity.

4.5 Implications and Limitations

These findings have crucial implications for penalty shootout design. The empirical evidence suggests that transitioning from ABAB to ABBA is the most effective fairness-enhancing step. Implementing Catch-Up or Behind-First sequences, as proposed in Csató et al. (2021), would likely **introduce a new unfairness rather than correcting the existing one**, and furthermore it could be too big of a learning curve for the fans and also players.

However, there are limitations to my approach. While Bayesian updating provides a more realistic model of evolving probability estimates, it still relies on assumptions about prior distributions. Future research should validate these results using **actual penalty shootout datasets**, which means the players have to try all the penalty shootout sequence, rather than simulating scenarios, to determine whether the biases observed in the study persist in real-world conditions.

5 Summary and Discussion

5.1 Summary of Findings

This study examines the fairness of various penalty shootout sequences by integrating empirical data, bootstrapping techniques, and Bayesian updating. The research builds upon prior work by Csató et al. (2021) and Rudi et al. (2021), addressing the limitations of both by incorporating dynamic sequences and real-time probability updates.

In the **Introduction** chapter, the problem of fairness in penalty shootouts is introduced, focusing on the concept of First Mover Advantage (FMA). The chapter highlights the mathematical models developed by Csató et al. (2021) to analyze the impact of different penalty shootout mechanisms and their claim that dynamic sequences improve fairness.

In the **Problem Statement** chapter, gaps in previous research are identified. The study by Csató et al. assumes constant scoring probabilities throughout the shootout, ignoring real-world dynamics, while Rudi et al. improve upon this by using empirical data but only analyze static sequences. To address these shortcomings, this study extends the bootstrapping framework of Rudi et al. to include dynamic sequences while introducing Bayesian updating for a more realistic representation of scoring probabilities.

In the **Data Description and Quality** chapter, the datasets used in this study are detailed, including historical penalty shootout data and bootstrapped simulations. The distinction between **APH** datasets (aligned with Apesteguia and Palacios-Huerta, 2010) and **Extended** datasets, which include a broader range of competitions, is explained. The data quality assessment confirms the reliability of the datasets, with no missing values detected.

In the **Statistical Methods** chapter, the key methodologies used in the analysis are presented. Bayesian updating with a Beta-Binomial model is applied to dynamically adjust scoring probabilities after each kick, addressing the limitations of fixed probabilities in previous studies. Bootstrapping is employed to generate synthetic datasets, providing robust estimates of winning probabilities. The chapter also details the software tools used, including R packages such as `dplyr`, `plyr`, `sqldf`, and `markovchain`.

In the **Results and Discussion** chapter, empirical findings are presented and compared with theoretical predictions. The results confirm that ABAB remains biased in favor of Team A, while ABBA significantly reduces this bias. However, contrary to Csató et al.'s claims, dynamic sequences such as Catch-Up and Behind-First introduce a strong bias in favor of Team B, making them less fair than ABBA. The Thue-Morse (ABBA|BAAB) sequence also fails to achieve perfect fairness, slightly favoring Team B instead. These findings challenge prior assumptions and suggest that ABBA remains the most effective fairness-enhancing mechanism.

5.2 Discussion and Future Improvements

While this study offers important insights into penalty shootout fairness, several areas could be improved through future research.

1. Collecting More Granular Data: One limitation of this study is that it treats all penalty shootouts equally, without accounting for external factors such as match conditions, player fatigue, or goalkeeper skill level. Future studies could incorporate additional variables, such as weather conditions, stadium pressure, and player psychological profiles, to assess how these factors influence scoring probabilities dynamically.

2. Accounting for Player and Goalkeeper Skill Levels: Currently, the analysis assumes a uniform probability of scoring based on empirical averages. However, in reality, individual players have different penalty conversion rates, and goalkeepers have varying save percentages. Future research could incorporate individual player statistics, such as historical penalty success rates and goalkeeper diving tendencies, to refine scoring probability estimates and improve the Bayesian updating process.

3. Testing Dynamic Sequences in Real Matches: While the results suggest that Catch-Up and Behind-First sequences introduce a bias toward Team B, these findings are based on simulated data. A real-world implementation of these sequences, tested in professional competitions, could provide empirical validation of these effects. If FIFA or other governing bodies were to experiment with dynamic sequences in lower-tier tournaments, data from actual matches could help determine whether these biases persist in practice.

4. Extending Bayesian Updating with Hierarchical Models: The current Bayesia -n updating framework updates scoring probabilities based only on observed data within each shootout. A more advanced approach could use hierarchical Bayesian models, where priors are informed by historical data across multiple competitions, accounting for differences in scoring patterns across leagues and tournament formats.

5. Evaluating Alternative Fairness Criteria: This study primarily evaluates fairness based on the probability of Team A winning. However, other fairness criteria, such as psychological stress distribution or fan perceptions, could also be considered. Future research could explore alternative fairness definitions and assess how different penalty shootout sequences perform under these criteria.

6. Investigating the Impact of Sudden Death Rules: The sudden death stage in penalty shootouts introduces additional complexity, as the psychological pressure increases significantly. Future work could examine whether sudden death disproportionately affects the second shooter and whether modifications to the rules could mitigate this effect.

5.3 Concluding Remarks

This study advances the discussion on penalty shootout fairness by integrating empirical data, Bayesian updating, and bootstrapping techniques. The findings challenge prior claims that dynamic sequences improve fairness, demonstrating instead that these mechanisms systematically bias the shootout in favor of Team B. The results reaffirm the effectiveness of ABBA as a fairness-enhancing mechanism, while highlighting the need for further empirical validation of alternative shootout designs.

Future research should focus on incorporating more detailed contextual factors, testing dynamic sequences in real competitions, and refining statistical models to better capture the complexities of penalty shootouts. By continuing to refine these analyses, we can move closer to identifying the optimal penalty shootout format that ensures a truly fair contest for both teams.

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